

IBM and ARM to work together

IBM and ARM will be working on advanced semiconductor technologies designed to power the next generation of mobile devices

BlackBerry blinks finally

RIM revealed that it has made the BlackBerry Messenger service accessible to the government such that communication can now be intercepted or eavesdropped by the government

Intel Core i7-2600K and Core i5-2500K

The new generation Sandy Bridge processors are fast

Nimish Sawant

Sandy Bridge processors are based on Intel's high speed 32-nm fabrication process and is the second generation of the Core i3/i5/i7 processors. The two processors that we are testing are Intel Core i7-2600K and Intel Core i5-2500K.

This line of processors will need a completely new socket – LGA 1155. They are not backward compatible on the earlier LGA 1156 sockets even though the size of the socket is the same. However, you can still use the cooler on your LGA 1156 socket on the new motherboards. At present,

there are two chipsets namely Intel P67 and Intel H67 on which majority of the motherboards are based.

The Core i7-2600K is a quad-core 3.4 GHz processor supporting hyper-threading whereas Core i5-2500K is a quad-core 3.3GHz processor without hyper-threading. The K indicates that the processors have an unlocked multiplier.

These processors are based on the new Intel micro-architecture according to which the CPU cores, graphics core, memory controllers, power control unit, etc are on the same 32-nm die. Also all these blocks run on the same clock frequency of 100MHz.

Integrated Graphics is the crowning glory of this line of processors, as the graphics core is on the same die as the processor.

You can only overclock by adjusting the multiplier ie. Overclockers should only look at K series processors. Thanks to the integration of all the controllers like USB, SATA, power controllers, onto the same die as the cores, BCLK is set to 100 MHz for all blocks. So if you change the BCLK from 100MHz, it may mess up other controllers. Also H67 based motherboards will allow only graphics overclocking.

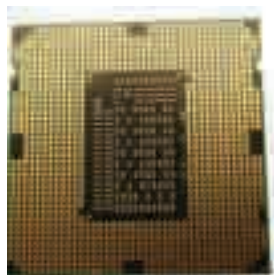
The Core i7-2600K scored better than the previous gen-

eration Core i7-875K in most of the tests. It performed 10% better in 3D Mark Vantage, 20% better in Cinebench R10 and POV Ray and around 15% better in HandBrake encoding.

In Core i5-2500K, the absence of hyperthreading was felt in some tests, but it still managed to beat the six-core AMD processors, which was quite good.

The Sandy Bridge processors are ideal for enthusiasts, who do not want to pay an arm and a leg for performance. Unlocked cores and a lower TDP enable you to easily overclock above 4.5 GHz on air cooling. Theoretically, the fully unlocked multipliers can be stretched upto 57x ie. 5.7GHz operating speed.

The Core i7-2600K is priced at ₹18,000 and Core i5-2500K is around ₹12,000. For those wanting to build a new mainstream rig Sandy Bridge processors are definitely recommended.



Intel Core i5-2500K



Intel Core i7-2600K

Contact

Intel

Phone: 91 80 2854 2105

Email: sasupport@mailbox.intel.com

Web site: www.intel.com

Scores and Ratings

Brand	Intel		Intel	
Model	Core i7 2600K		Core i5 2500K	
Price	₹ 18,000		₹ 12,000	
Turbo Boost Enabled	No	Yes	No	Yes
Core Frequency	3.4	3.4	3.3	3.3
Performance				
Synthetic Benchmarks				
3D Mark Vantage (Overall / CPU Score)	15537 / 53023	15474 / 53744	15457 / 50846	15432 / 52466
Cinebench R10 CPU Rendering (800 x 600, all cores)	18187	18834	15428	16008
Real World Benchmarks				
Maya 8.0 Rendering (sec) (1K pixels, prod quality) all cores	9	9	11	10
HandBrake x264 encoding (sec) 300 MB VOB to MP4	122	118	150	147
iTunes (converting 180 MB songs from MP3 to AAC) (secs)	110	106	110	112
Far Cry 2 (1280x1024, DX10, High)	115	115	115.9	115
Crysis Warhead (1280x1024, DX10, Med)	74	75	74	75
Ratings				
Features	7.5		7	
Performance	8.5		7	
Value for money	7		7.5	
Overall	7.5		7	

For a more detailed review on the Sandy Bridge processors log on to <http://bit.ly/gNWFz0>